



Review article

Human epidemiological evidence about the association between air pollution exposure and gestational diabetes mellitus: Systematic review and meta-analysis

Cheng-Yang Hu^{a,b,1}, Xiang Gao^{c,1}, Yuan Fang^d, Wen Jiang^b, Kai Huang^b, Xiao-Guo Hua^b,
Xiao-Jing Yang^b, Hong-Bo Chen^e, Zheng-Xuan Jiang^{c,*}, Xiu-Jun Zhang^{b,**}

^a Department of Humanistic Medicine, School of Humanistic Medicine, Anhui Medical University, 81# Meishan Road, Hefei, 230032, China

^b Department of Epidemiology and Biostatistics, School of Public Health, Anhui Medical University, 81# Meishan Road, Hefei, 230032, China

^c Department of Ophthalmology, The Second Affiliated Hospital of Anhui Medical University, 678# Furong Road, Hefei, 230601, China

^d Department of Public Health, Erasmus MC University Medical Center, P.O. Box 2040, 3000, CA, Rotterdam, the Netherlands

^e Department of Obstetrics and Gynecology, Maternal and Child Health Hospital Affiliated to Anhui Medical University, 15# Yimin Road, Hefei, 230001, China

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ABSTRACT

Background: Previous studies have shown that ambient air pollution exposure can increase the risk of type 2 diabetes mellitus (T2DM) significantly. In consideration of the common underlying pathophysiologic mechanisms, exposure to air pollution may also increase the risk of gestational diabetes mellitus (GDM), but the current evidence was inconsistent and has not well been systematically reviewed. Our goal was to perform a systematic review and meta-analysis assessing the association between air pollution exposure and GDM.

Methods: An extensive literature search was conducted in selected electronic databases for related human epidemiological studies published in English language. Summary effect estimates were calculated using random-effect models for a) risk per unit increase in continuous air pollutant concentration and b) risk of high versus low exposure level in individual study if each exposure that had been examined in ≥ 2 studies. We evaluated the heterogeneity using Cochran's Q test and quantified it by I^2 statistic. Publication bias was also evaluated through the funnel plot when sufficient number of studies are available.

Results: A total of 11 studies evaluating the association between GDM and exposure to air pollution were identified finally. The summary odds ratio (OR) for incidence of GDM following a $10 \mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ exposure during the second trimester was 1.04 (95% Confidence Interval (CI): 1.01, 1.09) and in NO_x during the first trimester was 1.03 (95%CI: 1.00, 1.07) per 10 ppb increase, while for high versus low SO_2 exposure during the second trimester was 1.25 (95%CI: 1.02, 1.53). High heterogeneity among study-specific results in majority of the analyses were observed, and attributed to different exposure assessment methods, populations, study locations, and covariates adjustment. Publication bias cannot be excluded because of the inclusion of small number of studies.

Conclusions: The present study supports the evidence that air pollution exposure increases the risk the GDM, albeit the existence of high heterogeneity. Further studies are necessary to elaborate the suggestive associations. These results are of public health significance since worsening air pollution in developing countries has been expected to increase the risk of GDM development.

1. Introduction

Gestational diabetes mellitus (GDM), defined as any degree of glucose intolerance with onset or firstly diagnosed during pregnancy (Hu et al., 2015), with the prevalence of 1%–14% worldwide, and has

continued to increase over the last couple of decades (Erem et al., 2015; Ferrara, 2007). The detrimental effects of GDM on both mothers and offspring have been well reported. For mothers who have GDM are more easily to develop preeclampsia during pregnancy and T2DM 5–10 years after delivery (Bellamy et al., 2009; Lai et al., 2016). For

* Corresponding author.

** Corresponding author.

E-mail addresses: jiangzhengxuan@ahmu.edu.cn (Z.-X. Jiang), zhangxiujun@ahmu.edu.cn, jzx1287@163.com (X.-J. Zhang).

¹ These authors contributed equally to this work.

offspring, a series of risks including malformations (Fadl et al., 2010), macrosomia (Metzger et al., 2008), diabetes (Clausen et al., 2008), cardiometabolic disease later in life (Spaight et al., 2016) are increased. Maternal characteristics including ethnicity, family history of T2DM, advanced age, and obesity have been shown to be related with the increased risk of GDM (Farahvar et al., 2019), but there are also up to half GDM cases without these classic determinants, which indicating the potential role for environmental factors (Fleisch et al., 2016). Identifying environmental triggers is necessary to women who are at risk and providing preventive opportunities.

Air pollution, as a remediable environmental trigger, was associated with the development of T2DM based on both human epidemiological and experimental evidence (Balti et al., 2014; Malmqvist et al., 2013; Sun et al., 2009). These evidences indicated that air pollution exposure influences oxidative stress, endothelial function, and inflammation which could results in insulin resistance, and all of these have been expected to be participated in the development of diabetes (Finch and Conklin, 2016). In consideration of the adverse effects of air pollution exposure upon diabetes among the general population, it is reasonable to propose that pregnant women exposure to air pollution may also increase the GDM risk by means of the common mechanisms. A growing body of human epidemiological evidence have indicated that pregnant women exposure to air pollution may be related with the increased risk of GDM (Choe et al., 2018, 2019; Fleisch et al., 2014, 2016; Hu et al., 2015; Malmqvist et al., 2013; Padula et al., 2019; Pan et al., 2017; Pedersen et al., 2017; Robledo et al., 2015; Shen et al., 2017). However, inconsistencies always exist regarding the critical exposure windows and effects of specific air pollutants. In consideration of the varied combinations of air pollutants and exposure windows (trimester/per-conception), and also the treatment of exposure as both a categorical or continuous variable, a systematic review and meta-analysis of these evidence is needed. Thus, we conduct the present study with 11 studies assessing the association between the two. We provide the summary effect estimates by different exposure windows, quantify heterogeneity across studies, evaluate publication bias using funnel plot when possible, and conduct sensitivity analyses.

2. Methods

2.1. Search strategies

An extensive search was done in the following bibliographic databases from their inception to May 27, 2019: PubMed, Medline, and Web of Science for relevant papers. We used the terms of “air pollutants”, “air pollution”, “particulate matter”, “NO₂”, “NO_x”, “PM_{2.5}”, “PM₁₀”, “SO₂”, “CO”, “O₃”, “nitrogen dioxide”, “ozone”, “sulfur dioxide”, “carbon monoxide”, “diabetes mellitus”, “diabetes”, “gestational diabetes”, “gestational diabetes mellitus”, “GDM” alone and in combination. No study designs were restricted. We also check the reference lists of the eligible studies to find the pertinent studies. After deduplication, two investigators independently check the eligibility of studies by screening the titles and abstracts and then potentially eligible studies were extracted as full texts. Any discrepancies were resolved by group discussion.

2.2. Inclusion and exclusion criterion

Study was included if it: 1) was original epidemiologic study reporting the relationship between air pollution exposure and GDM with effect estimates; 2) clearly defined the clinical outcome (GDM); 3) measuring exposure to air pollution of pregnant women using air pollutants concentrations; and 4) writing in the English language. Exclusion criterion included: 1) animal/review/letter studies; 2) studies on type 1 and type 2 diabetes mellitus; 3) studies only assessing whether diabetes status would modify the association between air pollution and birth outcomes; 4) studies with quantitative traffic density or

proximity to major roads data but without pollutant data. Two investigators independently reviewed the potential eligible studies, if there any discrepancies, they were resolved by group discussion.

2.3. Data extraction and quality assessment

Data extraction from eligible studies was also conducted independently by two investigators using a standardized data extraction sheet including the following information: 1) authors, publication year, and study period; 2) study design; 3) study location; 4) exposure (pollutant studied and concentration); 5) sample size; 6) exposure type (continuous or categorical); 7) covariates adjustment; 8) diagnostic criteria for GDM. We contacted the corresponding author of this study (Shen et al., 2017) in case of missing information of air pollutants concentrations. Effect estimates were extracted from the fully adjusted models. Newcastle-Ottawa Scale (NOS) was used to evaluate the quality of included studies separately for cohort and case-control studies. The NOS is a nine-point scale that assigns points on the basis of the process of selection of the cohorts or of the cases/controls (0–4 points), of the comparability of the cohorts or of the cases/controls (0–2 points), and of the identification of the exposure or of the outcomes of study participants (0–3 points). A study with score ≥ 7 , 4–6, or ≤ 3 was regarded as a high, intermediate, or low quality, respectively (Flores-Pajot et al., 2016).

2.4. Statistical analysis

Effect estimates were pooled if at least two studies reported the same pollutant and exposure window group available in the meta-analyses. We calculated two types of pooled effect estimates in the present meta-analyses: 1) for GDM risk per unit increase in continuous pollutant concentration and 2) for the comparison of GDM risk at high exposure level versus low exposure level in each study. If the included studies reported the effect estimates either by categorical exposure (quartiles, quintiles, tertiles) or for continuous exposure, we performed all the analyses using both categorical exposure defined as high versus low, where high level was reported as the highest study-specific category and low level was the lowest study-specific category and continuous exposure (if it's available). This method has been used previously in studies of environmental exposures (Kasdagli et al., 2019; Lin et al., 2013; Pedersen et al., 2014). If studies only presented the effect estimates for continuous exposure, they were standardized to the effect estimate per interquartile increase in exposure when the interquartile ranges (IQRs) were given, because these were considered to be comparable with effect estimates from the categorical studies (Gascon et al., 2016; Vrijheid et al., 2011).

In order to enable comparison across different studies, effect estimates were grouped by exposure windows (trimesters) and were harmonized to effect estimate per 10 $\mu\text{g}/\text{m}^3$ increase in PM_{2.5} and PM₁₀, 10 ppb increase in NO_x, NO₂, NO, and O₃, 5 ppb increase in SO₂, and 1 ppm increase in CO when exposure was considered as continuous variable. If exposure estimations were expressed in a mass per volume unit (e.g., mg/m^3) instead of ppm, ppb, or pphm were converted to ppb using the method provided by Malmqvist et al. (2013).

We retrieved the effect estimates (usually OR) and its 95% CI from the fully-adjusted models reported in each study. If both multiple and single-pollutant models were reported, we extracted the effect estimates from single-pollutant model.

Random effects models were applied to obtain the pooled effect estimates as the anticipated inconsistency in the studies because of the differences in study design, location, and population. Illustrative forest plots were used when one air pollutant combined with two or more exposure windows and each combination including at least three studies. The pooled ORs were also presented for pollutants by different exposure windows. We assessed the heterogeneity using the Cochran's Q test and the I² metric for each pollutant-outcome analysis in effect

estimate combined. To investigate the possible sources of heterogeneity, we would carry out subgroup analyses based on the factors like study design where feasible.

To test the robustness of results, we conducted several sensitivity analyses. In the study of Fleisch et al. (2014), they assessed air pollution exposure using assigned daily central-site values to women who lives within 40 km radius of the monitor and also using spatiotemporal models to estimate air pollutant concentration at each woman's residential address. We used the extracted effect estimates from spatiotemporal models as the main analysis, and results from the method of central-site values assignment to be sensitivity analyses. Meanwhile, Pedersen et al. (2017) used the criterions of WHO and Danish for GDM ascertainment, and reported different effect estimates. The results from WHO criterion were used as the main analysis and the Danish criterion was used as sensitivity analysis. We used funnel plot to investigate possible publication bias when 5 or more studies were available (Goldstein et al., 2017). STATA version 15.0 (StataCorp LP, College Station, Texas, USA) software was used for all analyses.

3. Results

3.1. Search results and study characteristics

A total of 979 studies were identified initially using our search strategy and 948 studies were excluded after the duplicates removed and initial screening. The remaining 31 studies were further screened for the abstracts and we excluded 13 studies based on different reasons. Subsequently, 18 full text studies were retrieved for detailed evaluation. Of these articles, 3 studies used traffic-related air pollution proxies including traffic density or proximity to major roadway, 3 studies assessed the outcome of glucose homeostasis/metabolism, 1 study provided the results from the same cohort. Consequently, the remaining 11 studies that reported the association between air pollution exposure and GDM with presented effect estimates were finally included in our meta-analysis (Fig. 1). The main characteristics of the included studies are presented in Table 1. Almost all of the studies were cohort study (Choe et al., 2018, 2019; Fleisch et al., 2016; Hu et al., 2015; Malmqvist et al., 2013; Padula et al., 2019; Pan et al., 2017; Pedersen et al., 2017; Robledo et al., 2015), except one study was case-control study design (Shen et al., 2017). Eight studies were conducted in USA (Choe et al., 2018, 2019; Fleisch et al., 2014, 2016; Hu et al., 2015; Pan et al., 2017; Robledo et al., 2015), 2 in Taiwan (Pan et al., 2017; Shen et al., 2017), 1 in Denmark (Pedersen et al., 2017), and 1 in Sweden (Malmqvist et al., 2013). 4 studies treated the air pollutants as both continuous and categorical exposure (Choe et al., 2019; Fleisch et al., 2014, 2016; Pedersen et al., 2017), 5 studies treated the exposure as continuous (Choe et al., 2018; Hu et al., 2015; Pan et al., 2017; Robledo et al., 2015; Shen et al., 2017), and 2 studies treated the exposure as categorical (Malmqvist et al., 2013; Padula et al., 2019). All the included studies assess the outcome using medical records or birth certificates, except one study using the self-reported outcome (Pan et al., 2017). Also, there was a large difference in the adjustment of covariates and variability in the sample sizes. The results of the quality assessment and the details of rating were summarized in Table S1. The quality scores of included studies ranged from 8 to 9 and all the studies are rated as high quality (NOS $\geq$ 7).

3.2. Outcome definition and prevalence

The definitions of outcome were relative similar among the studies (Table 1). Four studies (Fleisch et al., 2014, 2016; Hu et al., 2015; Pan et al., 2017) using the American Diabetes Association criteria for GDM diagnosis: pregnant women were classified as having GDM if they had two of the following abnormal values on the oral glucose tolerance test (OGTT): blood glucose > 95 mg/dL at baseline, > 180 mg/dL at 1 h, > 155 mg/dL at 2 h, or > 140 mg/dL at 3 h. Notably, Pan et al.

(2017) used the question to indirectly determine the prevalence of GDM among the pregnant women: "Have you been diagnosed as having gestational diabetes by your obstetrician during this pregnancy?", regardless of GDM was diagnosed by obstetrician based on the American Diabetes Association criteria as they reported. Padula et al. Choe et al., Shen et al. and Robledo et al. (Choe et al., 2018, 2019; Padula et al., 2019; Robledo et al., 2015; Shen et al., 2017) identified GDM cases from administrative data. In the study of Malmqvist et al. (2013), GDM was diagnosed based on plasma glucose > 10 mmol/L 2 h after oral administration of 75 g of glucose. Pedersen et al. (2017) used two types of diagnostic criteria, the Danish criterion: For venous plasma, the OGTT was considered abnormal (GDM) if two or more glucose values exceeded 6.2 mmol/L at 0 min, 10.9 mmol/L at 30 min, 11.1 mmol/L at 60 min, 9.2 mmol/L at 90 min, 8.9 mmol/L at 120 min, 8.2 mmol/L at 150 min, and 7.3 mmol/L at 180 min, and the WHO criterion: 75 g OGTT with either a fasting venous plasma glucose equal to or > 7.0 mmol/L and/or a 120 min value equal to or > 7.8 mmol/L. The reported prevalence for GDM ranged from 0.78% (Pedersen et al., 2017) to 8.0% (Choe et al., 2018).

3.3. Air pollution exposure

Almost all the included using routinely available air-monitored data as measurement basis and then assigned to maternal residential address at the time of delivery using environmental models (land-use regressions, satellite remote sensing or dispersion), whereas Shen et al. and Pan et al. (Pan et al., 2017; Shen et al., 2017) assigned exposure to maternal residential address at city/township level. Pedersen et al., (2017) have moved forward to considered the residential mobility and collected the information on residential address history during pregnancy. Exposure indices were usually calculated from pollutant measurements at the nearest monitoring station or as a distance-weighted average of measurements of all stations in the area. Distance from the monitors to maternal residential address as well as the number and density of monitors are differed across studies. Eight studies evaluated $PM_{2.5}$, 5 NO_2 , 4 O_3 , 3 PM_{10} , NO_x , or CO (Table 1). The distribution of pregnancy air pollution exposure varied across the included studies.

3.4. Combined effect estimates of air pollution and the risk of GDM

The summary effect estimates using either a categorical or continuous exposure were presented in Table 2, although there is a small number of included studies assessed the associations for several pollutant-exposure window groups. Regarding exposure to $PM_{2.5}$ and PM_{10} , none of the associations were observed for PM_{10} , but exposure to high versus low $PM_{2.5}$ during the first trimester reached a borderline level of statistical significance (OR: 1.04; 95% CI: 0.99, 1.10), exposure to continuous $PM_{2.5}$ during the second trimester was associated with higher odds of GDM (OR: 1.14; 95% CI: 1.01, 1.29, per $10 \mu g/m^3$). The summary effect estimates for the most other pollutants (NO_2 , O_3 , NO , CO , NO_x , SO_2) did not show any association. The analyses in investigating categorical or continuous exposures presented similar results for majority of pollutants. For exposure to high versus low SO_2 , the summary OR was 1.25 (95% CI: 1.02, 1.53) during the second trimester. Exposure to NO_x during the first trimester was also associated with higher odds of GDM, the summary OR was 1.03 (95% CI: 1.00, 1.07) per 10 ppb increase in NO_x concentration. Evidence of mild to high heterogeneity was found in all associations when assessing continuous and categorical exposure metrics. Supplementary Figs. S1–S6 present the forest plots for the associations between air pollutants and GDM when one air pollutant combined with two or more exposure windows and each combination including at least three studies were available, namely for the analysis of continuous and categorical $PM_{2.5}$, NO_2 , and O_3 exposure. In the present study, we didn't perform subgroup analysis to assess the heterogeneity because of the small number of studies included.

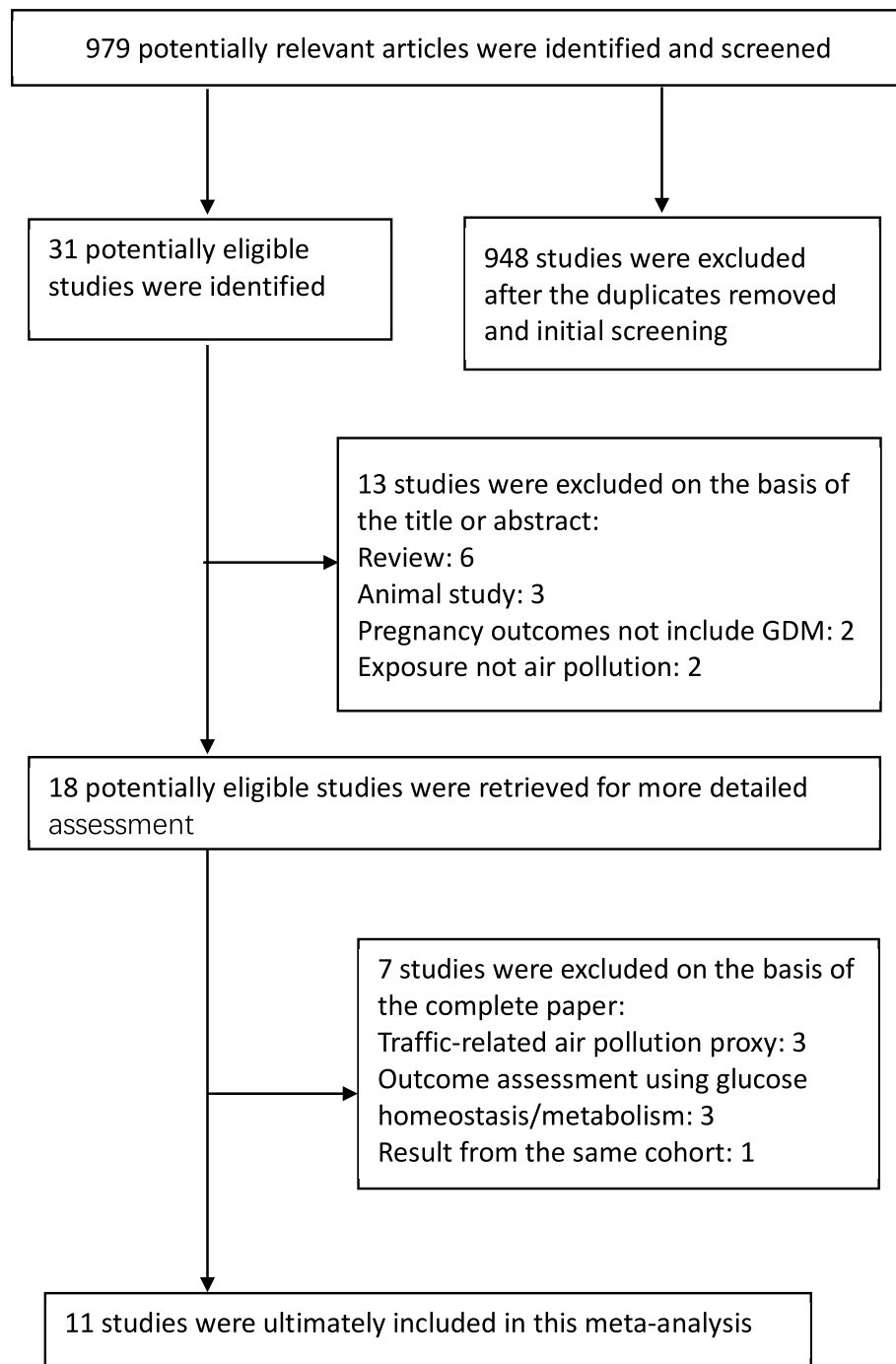


Figure 1. Flowchart for the systematic literature search on air pollution exposure and risk of gestational diabetes mellitus.

3.5. Sensitivity analyses

For the sensitivity analysis, we replaced the effect estimates for continuous exposure to $PM_{2.5}$ in the second trimester and NO_2 exposure in the first, second, and third trimesters. All these sensitivity analyses reported similar results, which indicating the robust of results.

3.6. Publication bias

The small number of included studies prohibits us to assess the publication bias but nevertheless the funnel plots were used when five or more studies were available, and relative low risk of publication biases were found for continuous and categorical $PM_{2.5}$ exposure during

the first trimester. The results of the funnel plots for continuous and categorical exposures to $PM_{2.5}$ during the first trimester were not symmetrical, indicating the existence of publication bias (Supplementary Figs. S7–S10).

4. Discussion

We conducted this meta-analysis of studies evaluating the association between air pollution exposure and GDM. Our study suggested indications of associations with continuous exposure to $PM_{2.5}$ and NO_x during the first trimester and also for high versus low SO_2 exposure during the second trimester. It remains unclear whether the inconsistent trimester-specific results reflect true differences in the etiology

Table 1
Characteristics of selected epidemiological studies investigating the association between air pollution exposure and gestational diabetes mellitus.

References (time period)	Design	Location	Pollutant and concentration	Sample size	Exposure type	Covariates adjusted	Diagnostic criteria for GDM
Fleisch et al. (2016) 2003–2008	Cohort	Massachusetts, USA	PM _{2.5} [µg/m ³ , Mean (SD)] T1: 10.4 (1.7), T2: 10.4 (1.7)	159,373	Continuous, categorical	Maternal characteristics (age, smoking habits, race/ethnicity, education, prenatal insurance), census tract characteristics (median household income, percent open space, and median value of owner occupied housing), and timing of birth (season and date)	≥ 2 of the following abnormal values on the OGTT: blood glucose > 95 mg/dL at baseline, > 180 mg/dL at 1 h, > 155 mg/dL at 2 h, or > 140 mg/dL at 3 h
Choe et al. (2018) 2002–2012	Cohort	Rhode Island, USA	PM _{2.5} [µg/m ³ , Mean (SD)] T1: 9.7 (1.9), T2: 9.6 (1.9), T3: 9.5 (2.1) BS [µg/m ³ , Mean (SD)] T1: 0.5 (0.1), T2: 0.5 (0.1), T3: 0.5 (0.1)	61,640	Continuous	Mother's age, tobacco use, parity, education, race, insurance, marital status, NSES z-score, year of last menstrual period, and conditional on town of residence (39 towns total)	NA: from birth certificate data
Hu et al. (2015) 2004–2005	Cohort	Florida, USA	PM _{2.5} [µg/m ³ , Median (IQR)] T1: 9.65 (2.61), T2: 9.76 (2.61), Full pregnancy: 9.91 (2.02) O ₃ [ppb, Median (IQR)]: T1: 36.48 (7.83), T2: 36.95 (8.00), Full pregnancy: 37.84 (7.09) NO ₂ [µg/m ³ , Median (p5–p95)] T1: 11.5 (5.8, 27.4)	410,267	Continuous	Maternal age, race, education, marital status, season of conception, year of conception, prenatal care began, urbanization, and median household income	≥ 2 of the following abnormal values on the OGTT: blood glucose > 95 mg/dL at baseline, > 180 mg/dL at 1 h, > 155 mg/dL at 2 h, or > 140 mg/dL at 3 h
Pedersen et al. (2017) 1997–2002	Cohort	Denmark	NO ₂ [µg/m ³ , Median (p5–p95)] T1: 11.5 (5.8, 27.4)	72,745	Continuous, categorical	Maternal age, parity, pre-pregnancy BMI, height, disposable income, education, and season of conception	≥ 2 of the following abnormal values on the OGTT: > 6.2 mmol/L at 0 min, > 10.9 mmol/L at 30 min, > 11.1 mmol/L at 60 min, > 9.2 mmol/L at 90 min, > 8.9 mmol/L at 120 min, > 8.2 mmol/L at 150 min, > and 7.3 mmol/L at 180 min
Pan et al. (2017) 2004–2005	Cohort	Taiwan	PM ₁₀ [µg/m ³ , Median (IQR)] T1: 55.1 (23.8), T2: 55.2 (21.8), T3: 56.2 (27.1) CO [ppm, Median (IQR)] T1: 0.6 (0.3), T2: 0.6 (0.3), T3: 0.5 (0.3) NO [ppb, Median (IQR)] T1: 6.0 (4.2), T2: 6.3 (4.1), T3: 6.2 (4.0) NO ₂ [ppb, Median (IQR)] T1: 20.4 (7.7), T2: 20.1 (8.1), T3: 18.9 (8.8) NO _x [ppb, Median (IQR)] T1: 26.8 (11.0), T2: 26.6 (11.1), T3: 25.4 (12.1) SO ₂ [ppb, Median (IQR)] T1: 4.1 (1.8), T2: 4.4 (1.9), T3: 4.5 (1.8) O ₃ [ppb, Median (IQR)] T1: 25.7 (5.1), T2: 25.4 (4.2), T3: 25.1 (5.2)	19,606	Continuous	Maternal age, BMI, weight gain, fetal gender, parity, and annual household income	≥ 2 of the following abnormal values on the OGTT: blood glucose > 95 mg/dL at baseline, > 180 mg/dL at 1 h, > 155 mg/dL at 2 h, or > 140 mg/dL at 3 h
Shen et al. (2017) 2006–2013	Case-control	Taiwan	PM _{2.5} [µg/m ³ , Median (IQR)] Pre-conception: 13.26, T1: 12.86, T2: 12.86, SO ₂ [ppb, Median (IQR)] Pre-conception: 1.62, T1: 1.60, T2: 1.58 O ₃ [ppb, Median (IQR)] Pre-conception: 6.22, T1: 6.28, T2: 6.26 NO ₂ [ppb, Median (IQR)] Pre-conception: 7.51, T1: 7.51, T2: 7.51 PM _{2.5} [µg/m ³ , Median (IQR)] Pre-conception: 11.71 (5.54), T1: 11.85 (5.28) PM ₁₀ [µg/m ³ , Median (IQR)] Pre-conception: 21.64 (6.30), T1: 21.62 (6.32) SO ₂ [ppb, Median (IQR)] Pre-conception: 3.36 (3.30), T1: 3.34 (3.31)	13,434	Continuous	Age at delivery, primipara, monthly income, urbanization, township-specific median family annual income quartiles, season of delivery, diagnosed comorbidity and charlson comorbidity index	NA: identified from Taiwan's National Health Insurance Research Data
Robledo et al. (2015) 2002–2008	Cohort	USA		219,952	Continuous	Maternal age, race, and study site	NA: identified from medical record

(continued on next page)

Table 1 (continued)

References (time period)	Design	Location	Pollutant and concentration	Sample size	Exposure type	Covariates adjusted	Diagnostic criteria for GDM
Fleisch et al. (2014) 1999–2002	Cohort	Massachusetts, USA	O ₃ [ppb, Median (IQR)]	2093	Continuous, categorical	Age, pre-pregnancy BMI, pregnancy weight gain, education, race/ethnicity, family history of diabetes, prior GDM, and season of last menstrual period	≥ 2 of the following abnormal values on the OGTT: blood glucose > 95 mg/dL at baseline, > 180 mg/dL at 1 h, > 155 mg/dL at 2 h, or > 140 mg/dL at 3 h
			Pre-conception: 29.71 (12.33), T1: 29.21 (12.36)				
			NO ₂ [ppb, Median (IQR)]				
			Pre-conception: 24.27 (28.55), T1: 25.18 (30.21)				
			CO [$\mu\text{g}/\text{m}^3$, Median (IQR)]				
Malmqvist et al. (2013) 1999–2005	Cohort	Sweden	Pre-conception: 0.54 (0.26), T1: 0.55 (0.26)	81,110	Categorical	Parity, BMI, maternal age, calendar year, and country of origin and exposure for second trimester.	Plasma glucose > 10 mmol/L 2 h after oral administration of 75 g of glucose
			Central-site PM _{2.5} [$\mu\text{g}/\text{m}^3$, Mean $\pm$ SD]				
			T2: 10.9 $\pm$ 1.4 (8.3–17.2)				
			Spatiotemporal PM _{2.5} [$\mu\text{g}/\text{m}^3$, Mean $\pm$ SD (range)]				
			T2: 11.9 $\pm$ 1.4 (8.5–15.9)				
Padula et al. (2019) 2000–2006	Cohort	San Joaquin Valley, USA	Central-site BC [$\mu\text{g}/\text{m}^3$, IQR] 0.16	252,205	Categorical	Maternal race, age, education, payment of birth expenses/insurance type, prenatal care (and for pollutants, the alternative trimester of exposure)	NA: identified from the hospital discharge records of the mother
			NO _x ($\mu\text{g}/\text{m}^3$)				
			Q1 = 2.5–8.9				
			Q2 = 9.0–14.1				
			Q3 = 14.2–22.6				
Choe et al. (2019) 2008–2010	Cohort	New York, USA	Q4 = > 22.7	256,372	Continuous, categorical	Maternal age, ethnicity, education, Medicaid status, parity, working status during pregnancy, conception year, deprivation index, and BMI, conditional on zip code of residence	NA: identified from discharge diagnoses
			PM _{2.5} [$\mu\text{g}/\text{m}^3$, Median (IQR)]				
			Entire pregnancy: 17.05 (6.39) CO [ppm, Median (IQR)] Entire pregnancy: 0.49 (0.18)				
			PM ₁₀ [$\mu\text{g}/\text{m}^3$, Median (IQR)]				
			Entire pregnancy: 35.93 (11.55)				

Abbreviations: SD, standard deviation; T1, the first trimester; T2, the second trimester; T3 the third trimester; BS, black smoke; NSES, neighborhood socioeconomic status; p5, 5th percentile; p95, 95th percentile; BMI, body mass index; IQR, interquartile range; BC, black carbon; GDM, gestational diabetes mellitus.

Table 2

Summary odds ratio estimates (and associated 95% confidence intervals (CI)) from random-effects meta-analysis for the association between air pollution exposure and gestational diabetes mellitus.

Pollutant and exposure window combination	Studies included ^a	Total number of cases ^b	Continuous exposure		High versus low exposure	
			Heterogeneity I ² (p-value)	Summary OR (95% CI)	Heterogeneity I ² (p-value)	Summary OR (95% CI)
PM _{2.5}						
Pre-conception	6, 7	18,061	82.3% (0.017)	1.01 (0.90, 1.14)	88.9% (0.003)	1.03 (0.91, 1.17)
Trimester 1	1, 2, 3, 6, 7, 10, 11	61,016	87.3% (0.000)	1.07 (0.94, 1.23)	84.9% (0.000)	1.03 (0.98, 1.08)
Trimester 2	1, 2, 3, 6, 8, 10, 11	60,807	82.7% (0.000)	1.14 (1.01, 1.29)	79.2% (0.000)	1.04 (0.99, 1.10)
Trimester 2 ^c	1, 2, 3, 6, 8, 10, 11	60,807	84.0% (0.000)	1.14 (1.00, 1.29)	82.1% (0.000)	1.03 (0.98, 1.09)
NO ₂						
Trimester 1	4, 5, 6, 10, 11	36,980	78.7% (0.003)	1.05 (0.92, 1.20)	85.1% (0.000)	1.01 (0.92, 1.12)
Trimester 1 ^d	4, 5, 6, 10, 11	37,335	76.0% (0.006)	0.98 (0.87, 1.11)	NA	NA
Trimester 2	4, 5, 6, 10, 11	36,980	67.9% (0.025)	1.04 (0.95, 1.14)	67.5% (0.026)	0.98 (0.94, 1.03)
Trimester 2 ^d	4, 5, 6, 10, 11	37,335	41.5% (0.163)	1.00 (0.94, 1.06)	NA	NA
Trimester 3	4, 5	538	82.4% (0.017)	1.24 (0.78, 1.98)	NA	1.00 (0.84, 1.19)
Trimester 3 ^d	4, 5	943	0.00% (0.587)	0.97 (0.82, 1.14)	NA	NA
O ₃						
Pre-conception	6, 7	18,051	70.3% (0.066)	0.97 (0.90, 1.05)	84.8% (0.010)	0.97 (0.89, 1.05)
Trimester 1	3, 5, 6, 7	32,461	95.7% (0.000)	1.02 (0.90, 1.16)	93.7% (0.000)	1.01 (0.93, 1.11)
Trimester 2	3, 5, 6	21,127	90.3% (0.000)	1.06 (0.87, 1.28)	94.5% (0.000)	1.04 (0.90, 1.21)
SO ₂						
Pre-conception	6, 7	18,051	99.0% (0.000)	1.68 (0.70, 4.04)	98.4% (0.000)	1.20 (0.92, 1.56)
Trimester 1	5, 6, 7	18,429	98.0% (0.000)	1.53 (0.82, 2.85)	97.0% (0.000)	1.16 (0.96, 1.40)
Trimester 2	5, 6	7095	94.6% (0.000)	1.94 (0.95, 3.96)	91.3% (0.001)	1.25 (1.02, 1.53)
PM ₁₀						
Trimester 1	5, 7, 10	24,322	0.0% (0.616)	0.98 (0.94, 1.02)	31.1% (0.234)	0.96 (0.93, 0.99)
CO						
Trimester 1	5, 7, 10	24,322	79.9% (0.026)	1.34 (0.61, 2.92)	85.4% (0.001)	0.99 (0.90, 1.09)
NO _x						
Trimester 1	5, 7	11,712	64.1% (0.095)	1.03 (1.00, 1.07)	70.6% (0.065)	1.12 (0.97, 1.30)
Trimester 2	5, 9	1971	NA	1.01 (1.00, 1.03)	86.8% (0.006)	1.31 (0.97, 1.75)

^a Studies included are different in each pollutant-exposure period meta-analysis depending on the data they published. References are as follows: 1, Fleisch et al. (2016); 2, Choe et al. (2018); 3, Hu et al. (2015); 4, Pedersen et al. (2017); 5, Pan et al. (2017); 6, Shen et al. (2017); 7, Robledo et al. (2015); 8, Fleisch et al. (2014). 9, Malmqvist et al. (2013); 10, Padula et al. (2019); 11, Choe et al. (2019).

^b Number of cases included in the continuous exposure analysis.

^c Sensitivity analysis using air pollutant data form Central-site instead of spatiotemporal.

^d Sensitivity analysis using the Danish criterion for gestational diabetes mellitus diagnosis instead of WHO criterion.

of disease versus random variation between studies, lack of information on the timing of GDM screening or diagnosis, differences in medical practice across locations, or other differences between studies. None associations were found with the other air pollutants.

The effect of air pollution exposure on chronic diseases including hypertension and diabetes mellitus have been well investigated by previous epidemiological studies (Clark et al., 2017; Fuks et al., 2017; Liu et al., 2017; O'Donovan et al., 2017; Pitchika et al., 2017; Renzi et al., 2018). Particularly, the association between GDM and air pollution exposure and GDM has also been investigated, but the current findings were inconsistent. Recently, Elshahidi published a meta-analysis regarding the association between outdoor air pollution exposure and GDM included eight studies (Elshahidi, 2019). Actually, one of the eight studies just evaluated the association between O₃ exposure and preterm birth among women who had developed GDM and Elshahidi improperly included this study (Lin et al., 2015). The author concluded that SO₂, NO, and NO_x exposures were associated with increased risk of GDM as he reported statistically significant summary effect estimates. But the summary effect estimates were from pooling the effect estimates that separately reported by trimester in original studies. In consideration of the relationship between air pollutants and GDM departures from linearity according to previous studies were reported (Malmqvist et al., 2013; Padula et al., 2019), hence in the present study we also compared effect estimates for high vs low pollutant exposures, which had not been evaluated by previous meta-analysis and verify the results using continuous exposures.

NO_x was always used as indicators of traffic-related air pollution and has been reported related to GDM in individual studies (Choe et al., 2018; Malmqvist et al., 2013; Robledo et al., 2015). In the Consortium on Safe Labor (CSL) study, Robledo et al. reported that per 30.21 ppb increment in NO_x exposure during the first trimester was related with an increased risk of GDM (OR = 1.06, 95%CI: 1.01, 1.10) (Robledo et al., 2015). A Swedish birth registry study reported that the highest quartile of NO_x comparing to the lowest quartile exposure increased the risk of GDM (OR = 1.69, 95%CI: 1.41, 2.03) during the second trimester (Malmqvist et al., 2013). In the Taiwan Birth Cohort Study, Pan et al. reported that per 1 ppb increase in NO_x exposure was related with a 2% increase in GDM risk during the first trimester (Pan et al., 2017). Notably, Pan et al. defined cases through self-reporting, which could not totally rule out the possibility of information errors. In contrast, the other two studies defined cases from medical records or discharge records.

Unlike NO_x and SO₂, which are mostly emitted from motor vehicles, PM_{2.5} is one of the most frequently assessed pollutants that released from multiple sources, including fossil-fuel combination by motor vehicles and stationary sources such as power plants (Shen et al., 2017). In the CSL study, Robledo et al. (2015) did not observe statistically significant association between PM_{2.5} exposure and GDM, but found that the nitrate of PM_{2.5} constituents increased the risk of GDM during the first trimester (OR = 1.05, 95%CI: 1.02, 1.09). Choe et al. (2018) and Hu et al. (2015) reported that PM_{2.5} exposure during the second trimester could increase the risk of GDM with the effect estimates of

1.08 (95%CI: 1.00, 1.15) and 1.15 (95% CI: 1.10, 1.20), respectively. In the study from Massachusetts Registry of Vital Records and Statistics, none of the trimester PM_{2.5} exposures were associated with increased risk of GDM in the full cohort, but the second trimester PM_{2.5} exposure was associated with higher odds of GDM (OR = 1.36, 95%CI: 1.08, 1.70) among women less than 20 years (Fleisch et al., 2016), and was also related with impaired glucose tolerance in another study (Fleisch et al., 2014). The observed association among subgroup of the population maybe due to the random chance or younger mothers are more likely to live in neighborhoods with higher levels of air pollution or less green space, or have less access to high quality prenatal care. Padula et al. (2019) reported non-significant associations in the first and second trimesters for high versus low PM_{2.5} exposures. In a newly published study, Choe et al. (2019) reported that the second trimester PM_{2.5} exposure was associated with an increased risk of GDM (OR = 1.06, 95%CI: 1.03, 1.10), but not during the first trimester, which was in accordance with our study. Also, the second trimester highest versus lowest quartiles of PM_{2.5} exposure was also associated with higher odds of GDM (OR = 1.10, 95%CI: 1.03, 1.18) (Choe et al., 2019). Shen et al. (2017) reported that exposure to PM_{2.5} (OR = 1.09, 95%CI: 1.02, 1.17; OR = 1.07, 95%CI: 1.01, 1.14) and SO₂ (OR = 1.37, 95%CI: 1.30, 1.45; OR = 1.38, 95%CI: 1.31, 1.46) during the first and the second trimesters were associated with significantly increased risk of GDM in Taiwan.

SO₂, similar to NO_x, was also mostly emitted from motor vehicles and manufacturing factories, and have been reported associated with increased risk of GDM previously. Except for the statistically significant association reported by Shen et al. (2017), Pan et al. (2017) reported that 1 ppb exposure in SO₂ increased the risk of GDM during the first trimester (OR = 1.05, 95%CI: 1.00, 1.11) and the second trimester (OR = 1.06, 95%CI: 1.00, 1.12). Robledo et al. (2015) reported that per 3.31 ppb increment in SO₂ exposure during the preconception and first trimester was associated effect estimates of 1.05 (95%CI: 1.01, 1.09) and 1.04 (95%CI: 1.00, 1.08), respectively. It seems that the association between SO₂ exposure and GDM was consistent. The overall inconsistent association between GDM and PM_{2.5} exposure shown in human observational studies is in contrast to an increasing number of epidemiologic studies showing relationship between PM_{2.5} and T2DM (Vrijheid et al., 2011).

Except for the criteria air pollutants that were widely used as air pollution exposure indices, several studies (Choe et al., 2018; Fleisch et al., 2014; Fleisch et al., 2016; van den Hooven et al., 2009; Yorifuji, 2015) also focused on proximity to major roadway or neighborhood traffic density, which were always treated as surrogates for traffic-related air pollution. Women living close to major roadway was associated with higher odds of GDM (OR = 1.09, 95%CI: 1.00, 1.19) in the study that conducted in a single hospital in Rhode Island (Choe et al., 2018). However, no associations were found between residential proximity to major roads and GDM in Japan (Yorifuji et al., 2015) and Netherlands (van den Hooven et al., 2009). On the other hand, living in a neighborhood with the highest (vs. the lowest) quartiles of traffic density was associated with higher odds of impaired glucose tolerance (OR = 2.66, 95% CI: 1.24, 5.71), but not associated with GDM in the Project Viva Cohort study (Fleisch et al., 2014). This finding was also supported by other studies that conducted in Massachusetts and Netherlands, none associations were detected between traffic density and GDM (Fleisch et al., 2016; van den Hooven et al., 2009). Actually, the power to detect a relationship between surrogate of air pollution exposure and GDM was lower because of traffic measures are crude estimations of air pollution, which could not assess the true level of personal exposure.

The underlying biological mechanisms of the possible association between air pollution and GDM are not fully understood presently. GDM likely shares pathways for development in common with T2DM in relation to air pollutants, since both of them are characterized by insulin resistance, and women with GDM are at increased risk of

developing T2DM after pregnancy (Xiang et al., 2011). Positive associations between air pollution, mainly PM_{2.5}, NO₂ or NO_x, and T2DM have been consistently reported (Eze et al., 2015; Liu et al., 2019). Therefore, many of the hypothesized mechanisms such as systematic inflammation, oxidative stress, and insulin resistance, which have been proposed to be involved in the development of diabetes may also apply to GDM (Eze et al., 2015).

Therefore, our finding of the association between PM_{2.5}, SO₂, and NO_x and GDM is likely true, but the possible inability to detect the effects of other pollutants cannot be completely excluded due to the following reasons. Firstly, misclassification of exposure due to the exposure assessment strategies including the use of residential address for exposure assessment, geocoding errors, or the difference between the true address during pregnancy and the address reported on the birth certificate, and the lack of information about time-activity patterns or indoor air pollution exposure which could have improved accuracy of exposure estimates. Similarly, the different methods like using administrative data for GDM ascertainment in the included studies, variation in clinical practice across study hospitals, or access to prenatal care across study participants may also introduce misclassification of outcome. Though we believe these biases would be non-differential and thus most likely bias the results towards the null. Secondly, all of the studies were conducted in developed countries/area with lower air pollution level, our null findings concerning other air pollutants may have been because of the relatively low average levels of air pollutants, in consideration of the limited statistical power to detect very small associations. Thirdly, we cannot exclude the possibility of uncontrolled confounders that might be related to pollution exposure and GDM risk, which could attenuate the associations. Therefore, these findings require replication in further studies.

Several limitations should be acknowledged in the present study. Firstly, significant heterogeneity was found among the most air pollutant-exposure window groups. Differences in the study sites, sample size, population demographics, individual characteristics, and analysis methods may account for the observed heterogeneity. Secondly, because of the relatively small number of included studies, as this research issue has attracted interest only past several years, we were unable to perform subgroup analyses, according the potential influent factors. Also, we extracted the effect estimates from single-pollutant models and then pooled them as only three studies reported multi-pollutant model effect estimates. The limited number of studies reported effects from multi-pollutant models hindering the deeply investigating of multi-pollutant exposure effect. But previous study (Kasdagli et al., 2019) that focused on similar research topic have reported that adjustment for other pollutants did not affect the pooled estimates. Meanwhile, the results from single-pollutant models remain robust when additional adjustment for the other pollutant in individual studies (Choe et al., 2019; Hu et al., 2015; Pan et al., 2017). Thirdly, all of the included studies are located in developed countries/areas, so when generalizing the results to developing countries like China and India with deadly air pollution (including indoor) requires attention. Despite the limitations, present study has indicated moderate associations between PM_{2.5}, SO₂, and NO_x exposure and the increased risk of GDM. Additional high-quality longitudinal studies, especially conducted in developing countries are needed (taking into consideration of potential covariates including air pollution sources and its compositions, noise exposure, spatial socioeconomic variables rather than individual socioeconomic characteristics, family history of diabetes, physical activity, and nutrition) is therefore needed to improve our understanding of this association.

5. Conclusions

This meta-analysis suggests that some air pollutants exposure during specific pregnancy trimesters may be related with the increased risk of GDM. Although the effect estimates were not high, the

population attributable fractions cannot be neglected given the ubiquitous presence of air pollution. Moreover, the findings need to be replicated in other studies considering the limited number of included studies. More studies with better study methodologies and exposure measurements are needed in the future.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envres.2019.108843>.

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